1 The equation of a curve is $y = kx^2 + kx + p$, where p and k are constants.

(a) Show that $p > \frac{k}{4}$ for which the curve lies completely above the x-axis [3]

(b) In the case where $k = 2$ and $p = 4$, find the values of $\boldsymbol{m}$ for which the line $y = mx - 4$ is a tangent to the curve. [4]